LAB # 1: INTRODUCTION TO MS ACCESS (TABLES, RELATIONS, FORMS)

Objective:

1-Introduction to MS ACCESS2-How to create tables.
3-How to create queries, reports, and forms.

Scope:

The student should know the following:

1. MS Access basics.
2. MS Access Practice.
3. Basic exercises.

Useful Concepts:

Table name, its column name and column's datatypes.How to create queries on tables.
Forms and Reports.

Discussion:

Queries, Reports, Forms.

Elements of databases

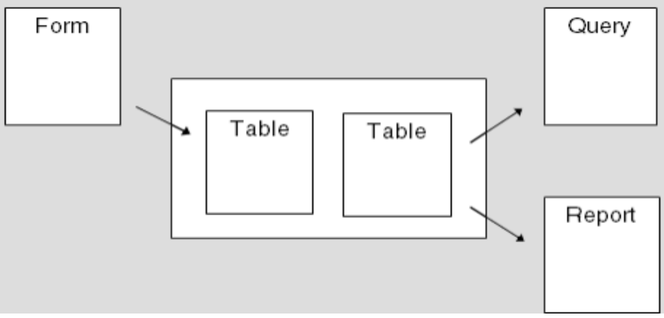
A database stores information in an organized way, and makes it easy to get information in and out.

Tables store data within the database.

Forms make it easy to put data into tables.

Queries pull out specific data.

Reports put data in an easily-read format.



Understanding Access Databases

As you already know, a database is a collection of information. In Access, every database is stored ina single file. That file contains *database objects*, which are the components of a database.
Database objects are the main players in an Access database. Altogether, you have sixdifferent types of database objects:

- **Tables** store information.
- **Queries** let you quickly perform an action on a table.
- **Forms** Forms provide an easy way to view or change the information in a table.
- **Reports** help you print some or all of the information in a table.

CREATE A BLANK NEW DATABASE:

1. **Start Access**.
   Here you can create a new database or open an existing one.

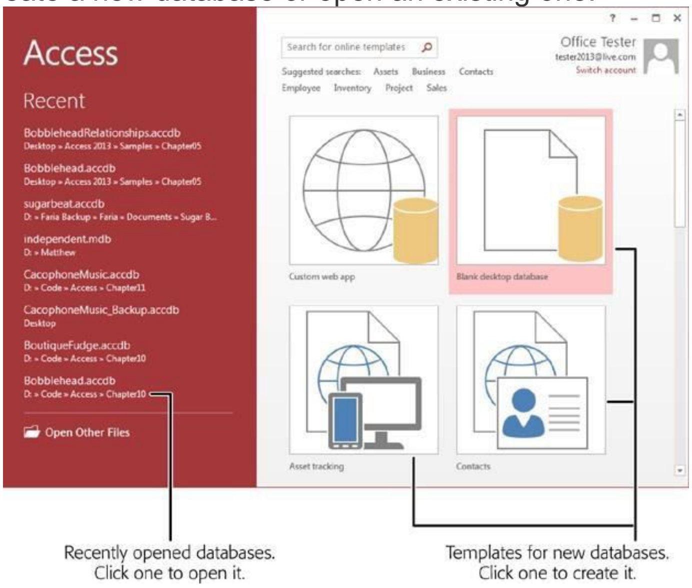


Figure 1-1. When you start Access, you see this two-part welcome page. On the left is a list of recently opened databases (if you have any). On the right is a list of templates that you can use to create a new database.

2. **Click the "Blank desktop database" template.**

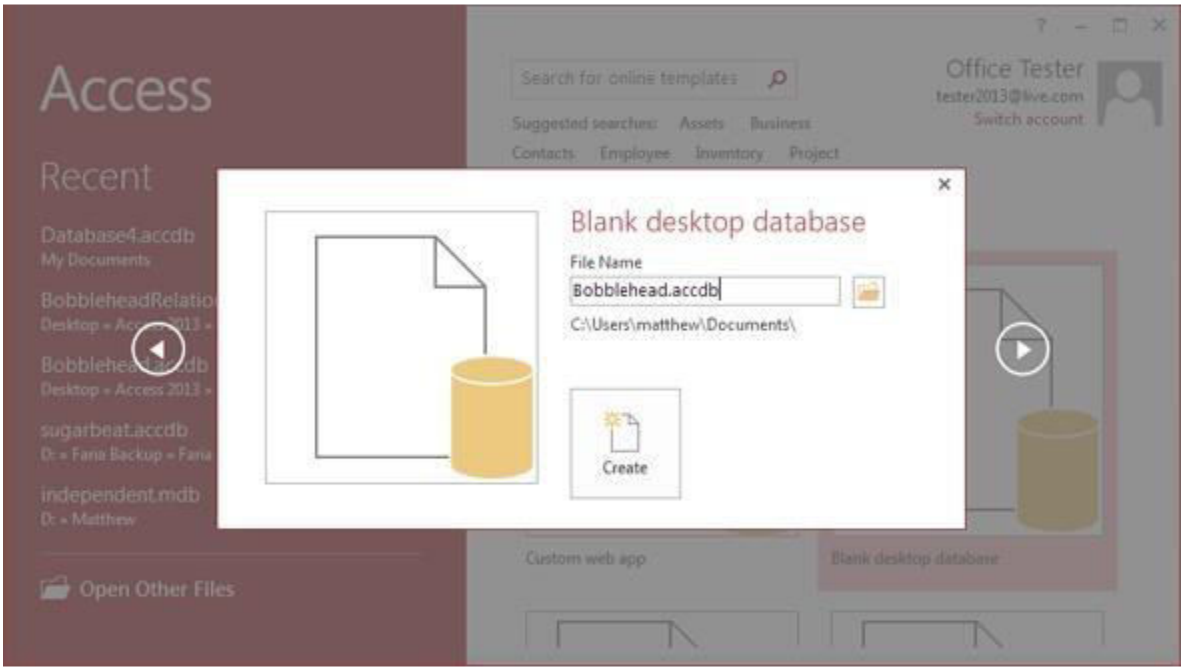
   When you choose to create a blank database, that is exactly what you get—a new, empty database file with no tables or other database objects.

   Other templates let you create databases that are preconfigured for specific scenarios and certain types of data. The cool sounding "Custom web app" template is a special case. It lets you create a web-enabled database that runs on SharePoint.

   No matter which template you click, Access pops open a new window that lets you choose a name and location for your new database.

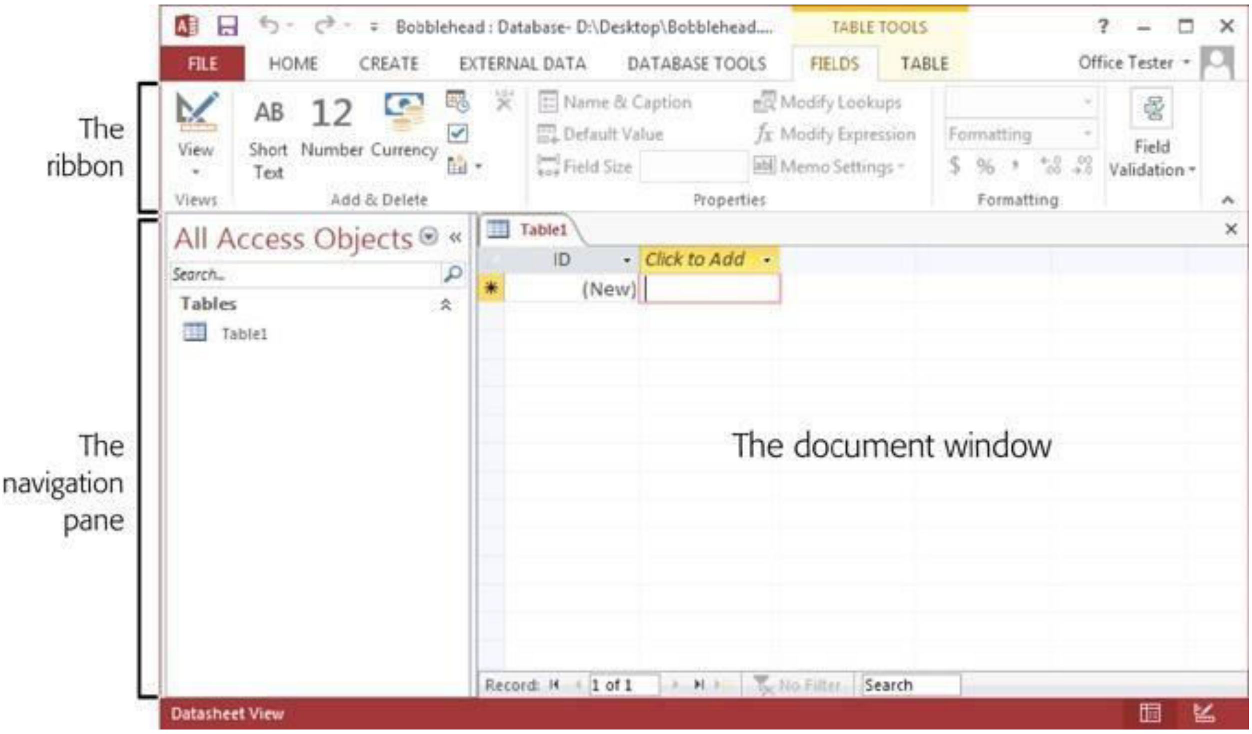
3. **Type a file name for the database you are about to create.**

   Access stores all the information for a database in a single file with theextension *.accdb* (which stands for "Access database").



4. **Click the big Create button (under the File Name box).**

Access creates your database file and then shows a datasheet where you can get towork creating your first table.



BUILDING YOUR FIRST TABLE

Tables are information containers. Every database needs at least one table—without it, you can't store any data. In a simple database, like the Bobblehead database, a single table (which we'll call
Dolls) is enough. But if you find yourself wanting to store several lists of related information, you needmore than one table.

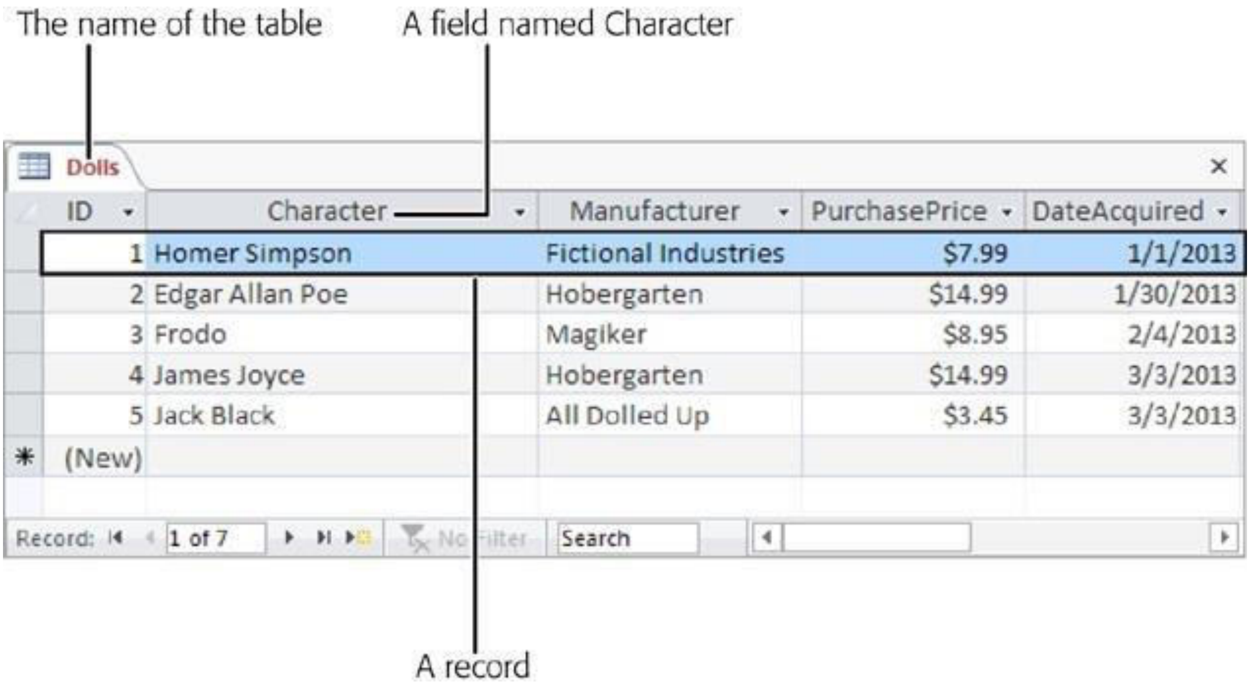


Figure 1-5. In a table, each record occupies a separate row. Each field is represented by a separate column. In this table, it's clear that you've added five bobblehead dolls. You're storing information for each doll in five fields (ID, Character, Manufacturer, PurchasePrice, and DateAcquired).

Before you start designing this table, you need to know some very basic rules:

- **A table is a group of** *records*. A record is a collection of information about a single thing. *In the Dolls table, for example, each record represents a single bobblehead doll. In aFamily table, each record would represent a single relative. In a Products table, each record*

  would represent an item that's for sale. You get the idea. When you create a new database,Access starts you out with a new table named Table1, although you can choose a more distinctive name when you decide to save it.
- **Each record is subdivided into** *fields*. Each field stores a distinct piece of information. *For example, in the Dolls table, one field stores the person on whom the doll isbased, another field stores the price, another field stores the date you bought it, and so on.*
- **Tables have a rigid structure**. In other words, you can't bend the rules. If you create four fields, *every* record must have four fields (although it's acceptable to leave some

fields blank if they don't apply).

- **Newly created tables get an ID field for free**. The ID field stores a unique number for each record. (Think of it as a reference number that will let you find a specific record later on.) The best part about the ID field is that you can ignore it when you're entering a new record. Access chooses a new ID number for you and inserts it in the record automatically.

Creating a Simple Table

When you first create a database, it's almost empty. But to get you started, Access creates your first database object—a table named Table1. The problem is, this table begins life completely blank, with no defined fields (and no data).

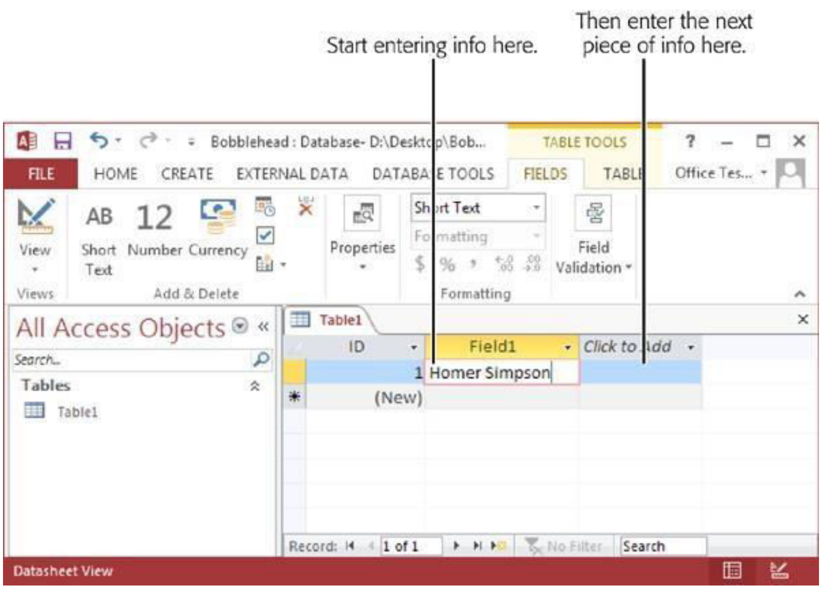
You can customize a table in two ways:

- **Design view** lets you precisely define all aspects of a table before you start using it.
- **Datasheet view** is where you enter data into a table. Datasheet view also lets you build a table on the fly as you insert new information.

<u>The following steps show you how to turn a blank new table (like Table1) into the Dolls table by using the Datasheet view:</u>

1. **To define your table, simply add your first record**.

2. **In the datasheet's rightmost column, under the "Click to Add" heading, type the first piece of information for the record (see Figure 1-6).**



3. **Press Tab to move to the next field, and return to step 2**

Note: *You may notice one quirk—a harmless one—when you add your first record. As you add new fields, Access may change the record's ID value of the record (changing it from 1 to 2 to 3, for example). Because the new record hasn't been inserted yet, every time you change the table's design by addinga new field, Access starts the process over and picks a new ID number, just to be safe. This automatic renumbering doesn't happen if you officially add the record (say, by moving down to the next row, or, in the ribbon, by clicking Home→Records→Save) and then add more fields to the table. However, there's really no reason to worry about the ID number. As long as it's unique—and Access guaranteesthat it is—the exact value is unimportant.*

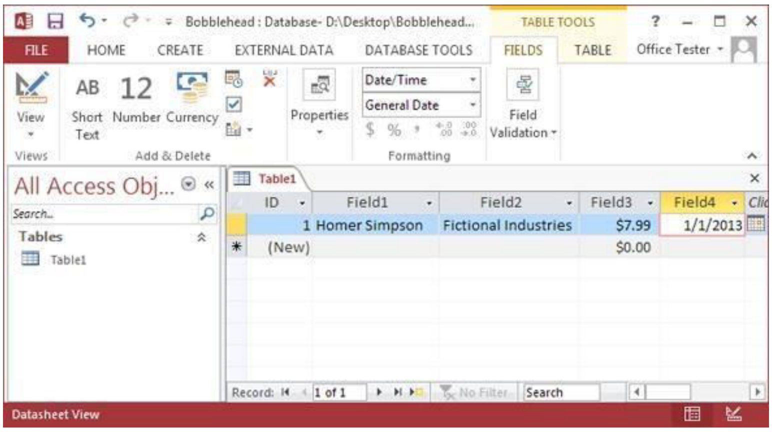


Figure 1-7. The only problem with this example so far is that as you enter a new record, Access creates spectacularly useless field names. You see its choices at the top of each column (they have names like Field1, Field2, Field3, and so on). The problem with using these meaningless names is that they may lead you to enter a piece of information in the wrong place. You could all too easily put the purchase price in the date column.

4. **It's time to fix your column names. Double-click the first column title (like Field1).**
5. **Type a new name, and then press Enter.**

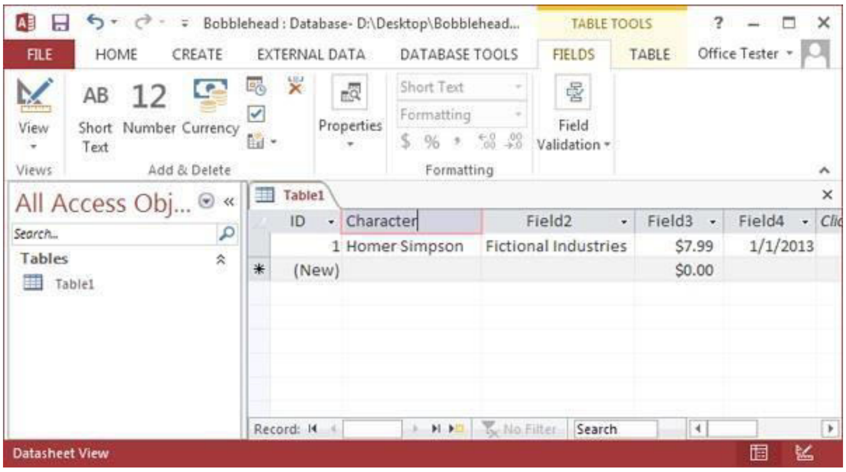


Figure 1-8. To specify better field names, double-click the column title. Next, type the real field name, and then press Enter. Page 90 has more about field naming, but for now just stick to short, text-only titles that don't include any spaces, as shown here.

6. **Press Ctrl+S or choose File→Save to save your table**.
7. **Type a suitable table name, and then click OK**.

Congratulations! The table is now a part of your database.

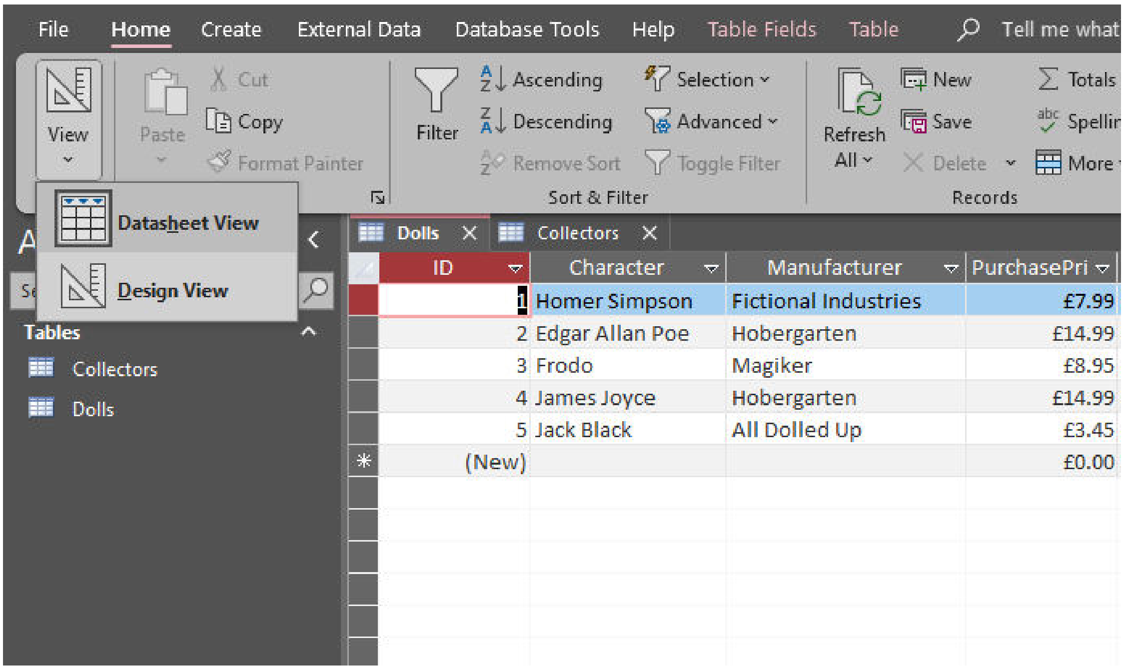
Editing a Table

To fill the Dolls table, you use the same datasheet you used to define the table. You can perform three basic tasks:

1. **Editing a record**. Move to the appropriate spot in the datasheet (using the arrowkeys or the mouse), and then type in a replacement value.
2. **Inserting a new record**. Move down to the bottom of the table to the row that has an asterisk (*) on the left. This row doesn't actually exist until you start typing some information. At that point, Access creates the row and moves the asterisk down to the next row. You can repeat this process endlessly to add as many rows as you want (Access can handle millions).
3. **Deleting a record**. You have several ways to remove a record, but the easiest is to right-click the margin immediately to the left of the record, and then choose Delete Record.

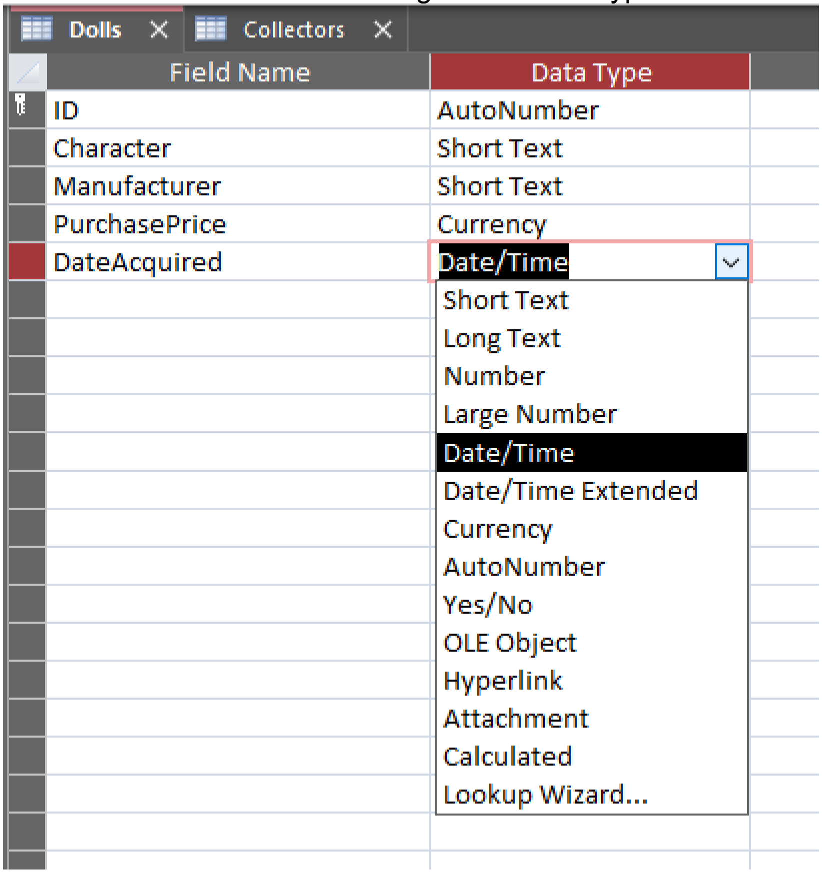
Adding Fields

Open the Design View from the bottom left of your screen or via the view column in the Home Section.

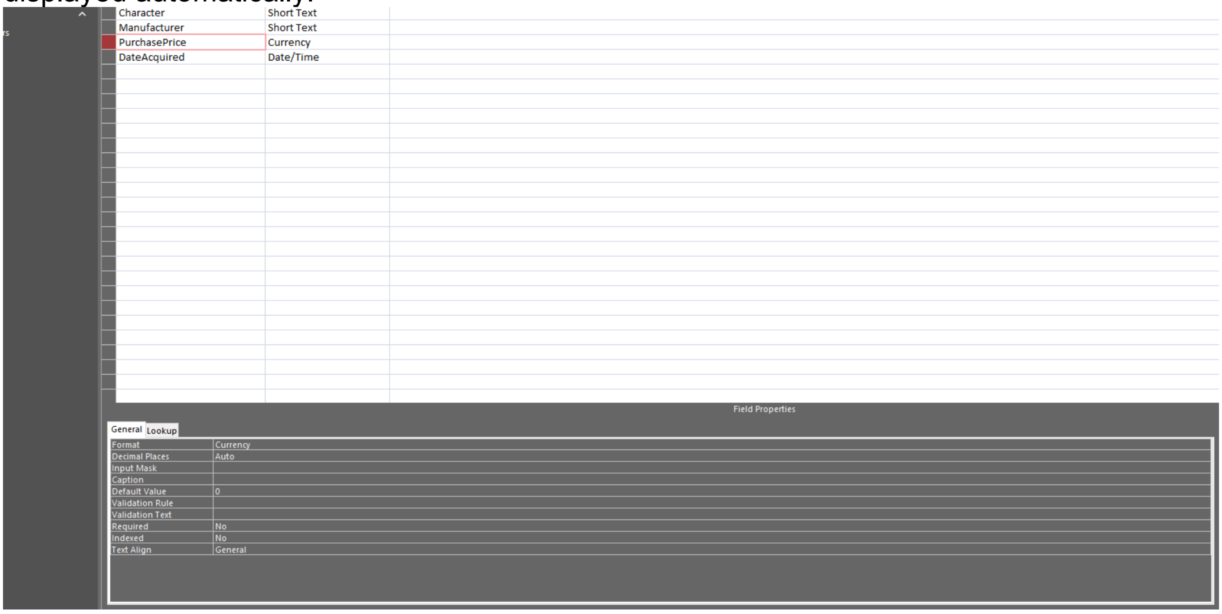


Design View

You can Add new fields or change their Data Type from the Design View Like so.



In order to edit a field's properties just left click on the desired field and its properties will be displayed automatically.



Datasheet Shortcut Keys

Power users know the fastest way to get work done is to use tricky keyboard combinations like Ctrl+Alt+Shift+*. Although you can't always easily remember these combinations, a couple of tables can help you out. Table 1-1 lists some useful keys that can help you whiz around the datasheet.

Table 1-1. Keys for Moving Around the Datasheet

KEY	RESULT
Tab (or Enter)	Moves the cursor one field to the right, or down when you reach the edge of the table. This key also turns off Edit mode if it's currently switched on.
Shift+Tab	Moves the cursor one field to the left, or up when you reach the edge of the table.This key also turns off Edit mode.
→	Moves the cursor one field to the right (in Normal mode), or down when you reachthe edge of the table. In Edit mode, this key moves the cursor through the text in the current field.
←	Moves the cursor one field to the left (in Normal mode), or up when you reach theedge of the table. In Edit mode, this key moves the cursor through the text in the current field.
↑	Moves the cursor up one row (unless you're already at the top of the table). Thiskey also turns off Edit mode.
↓	Moves the cursor down one row (or it moves you to the "new row" position ifyou're at the bottom of the table). This key also turns off Edit mode.
Home	Moves the cursor to the first field in the current row. This key brings you to beginning of the current field if you're in Edit mode.
End	Moves the cursor to the last field in the current row. This key brings you to the endof the current field if you're in Edit mode.
Page Down	Moves the cursor down one screenful (assuming you have a large table of information that doesn't all fit in the Access window at once). This key also turnsoff Edit mode.
Page Up	Moves the cursor up one screenful. This key also turns off Edit mode.
Ctr+Home	Moves the cursor to the first field in the first row. This key doesn't do anything ifyou're in Edit mode.
Ctrl+End	Moves the cursor to the last field in the last row. This key doesn't do anything ifyou're in Edit mode.

Table 1-2 lists some convenient keys for editing records.

Table 1-2. Keys for Editing Records

KEY	RESULT
Esc	Cancels any changes you've made in the current field. This key works only if you use it in Edit mode. Once you move to the next cell, the change is applied.(For additional cancellation control, try the Undo feature, described next.)
Ctrl+Z	Reverses the last edit. Unfortunately, the Undo feature in Access isn't nearly aspowerful as it is in other Office programs. For example, Access lets you reverseonly one change, and if you close the datasheet, you can't even do that. You can use Undo right after you insert a new record to remove it, but you can't usethe Undo feature to reverse a delete operation.
Ctrl+"	Copies a value from the field that's immediately above the current field. This trick is handy when you need to enter a batch of records with similar information. Figure 1-11 shows this often-overlooked trick in action.
Ctrl+;	Inserts today's date into the current field. The date format is based on computer settings, but expect to see something like "12-24-2013." You'll learn more about how Access works with dates on Date/Time.
Ctrl+Alt+S pace	Replaces whatever value you've entered with the field's default value. You'll learn how to designate a default value on Setting Default Values.

Cut, Copy, and Paste

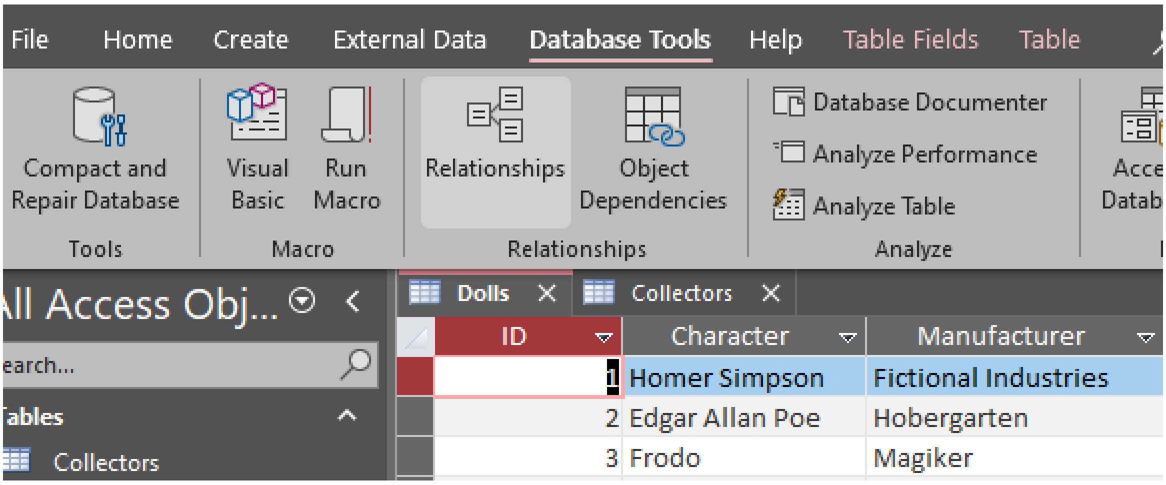
Access, like virtually every Windows program, lets you cut and paste bits of information from one spot to another. This trick is easy using just three shortcut keys: Ctrl+C to copy, Ctrl+X to cut (similar to copy, but the original content is deleted), and Ctrl+V to paste. When you're in Edit mode, you can use these keys to copy whatever you've selected. If you're not in Edit mode, the copying or cutting operation grabs all the content in the field.

Shrinking a Database

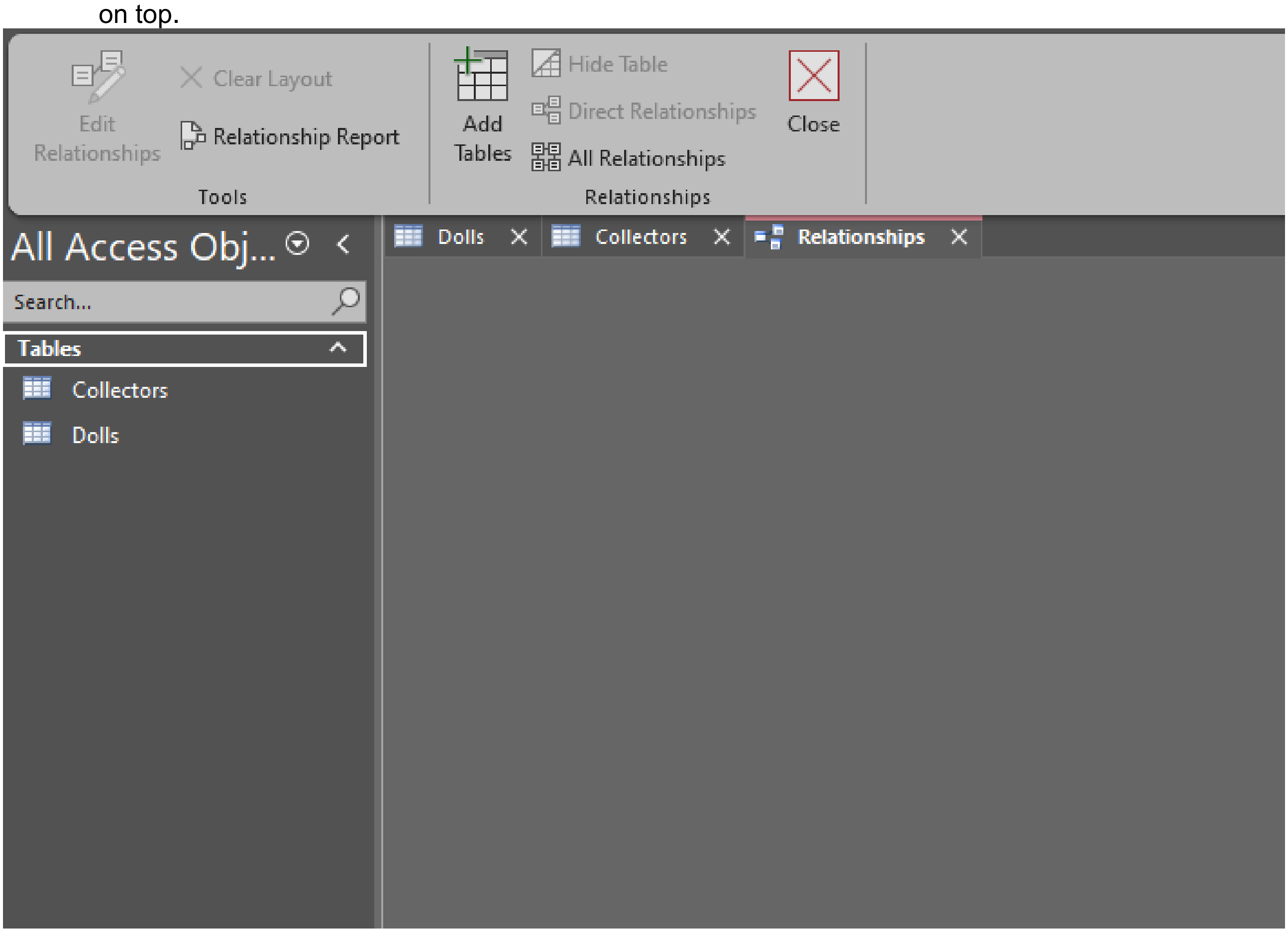
After you've been working with a database for a while, you might notice that its size bloats up like a week-old fish in the sun. If you want to trim your database back to size, you can use a feature called *compacting*. To do so, just choose File→Info and click the big Compact & Repair Database button. Access then closes your database, compacts it, and opens it again. If it's a small database, these three steps unfold in seconds. The amount of space you reclaim varies widely, but it's not uncommon to have a 20 MB database shrink down to a quarter of its size.

RELATIONSHIPS

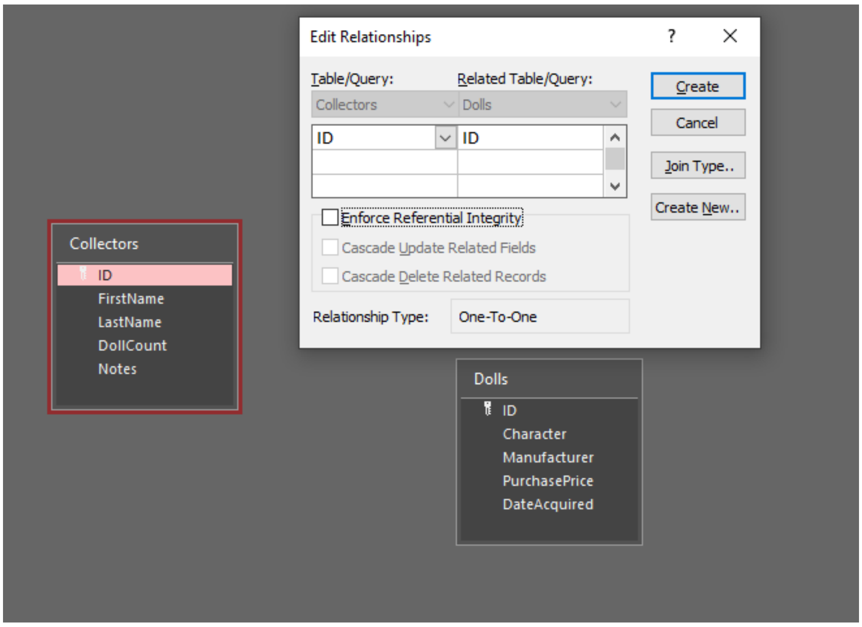
- Open the Relationships tab from the Database Tools Section.



- Now you can add the necessary tables you want to make relationships between either by drag and drop from the Shutter Bar on the left or you can or from the Add Tables from the header on top.



- To make a relationship between two fields you just have to left click on the first attribute and then drag on the second field you want to make the relationship with. After this a pop up will open through which you can chose which two fields to make the relationship between.



- Click on the Create button and you're done.

FORMS

There's a lot you can do design-wise with forms in Microsoft Access. You can createtwo basic types of forms −

- Bound forms
- Unbound forms

Bound Forms

Bound forms are connected to some underlying data source such as a table, query,or SQL statement.

Unbound Forms

- These forms are not connected to an underlying record or data source.
- Unbound forms could be dialog boxes, switch boards, or navigation forms.

Types of Bound Forms

There are many types of bound forms you can create in Access. Let us understand the types −

Single Item Form

This is the most popular one and this is where the records are displayed — one recordat a time.

Multiple Item Form

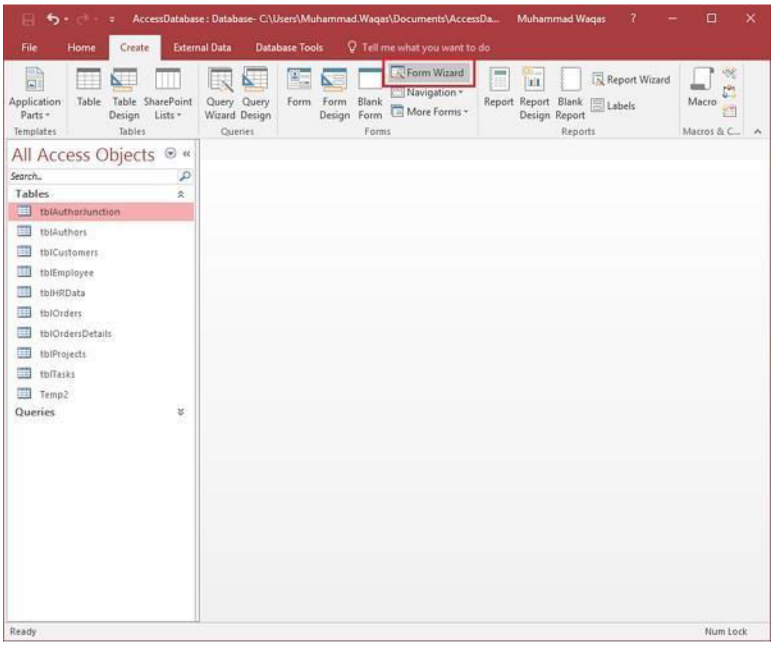
This displays multiple records at a time from that bound data source.

Split Form

The form is divided into halves, either vertically or horizontally. One half displays a single item or record, and the other half displays a list or provides a datasheet view of multiple records from the underlying data source.

Creating Forms

There are a few methods you can use to create forms in Access. For this, open yourDatabase and go to the **Create tab**. In the Forms group, in the upper right-hand corner you will see the Form Wizard button.

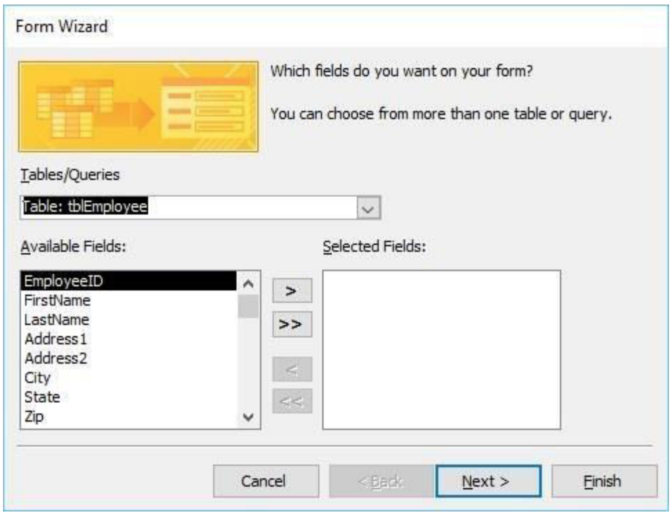


Click on that button to launch the Form Wizard.

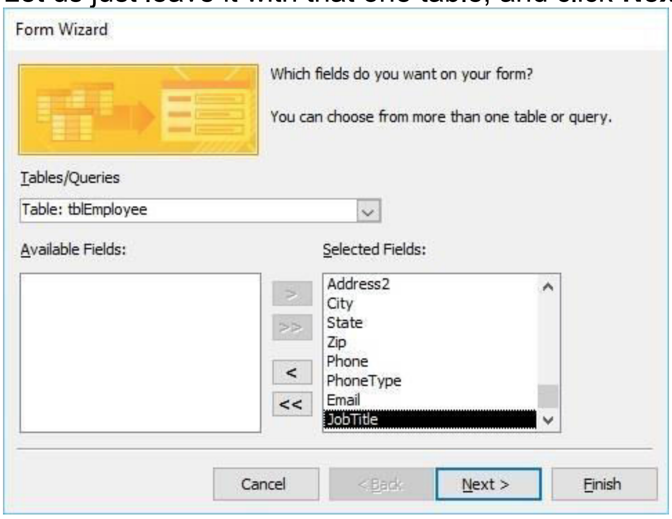
On this first screen in the wizard, you can select fields that you want to display onyour form, and you can choose from fields from more than one table or a query.

Let us assume we want to simply have a quick form that we are going to use for dataentry for our employee information.

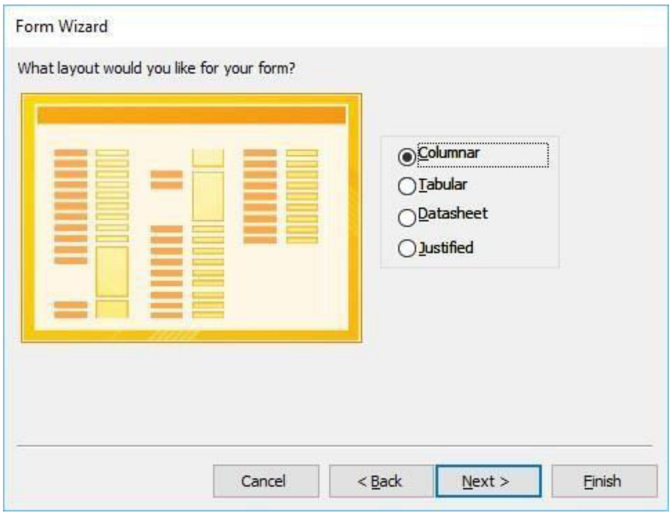
From **Tables/Queries** drop-down list, select **tblEmployees** table. Click on thedouble arrow to move all the fields at once.



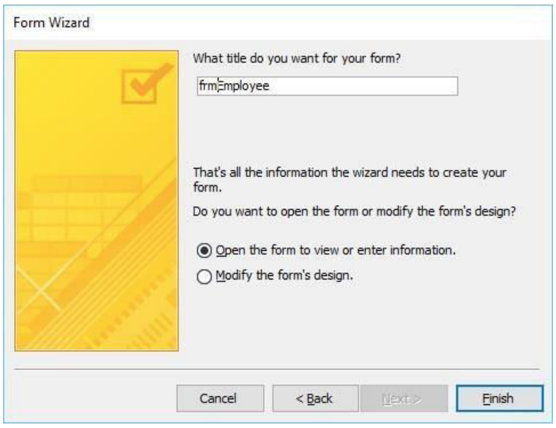
Let us just leave it with that one table, and click **Next**.



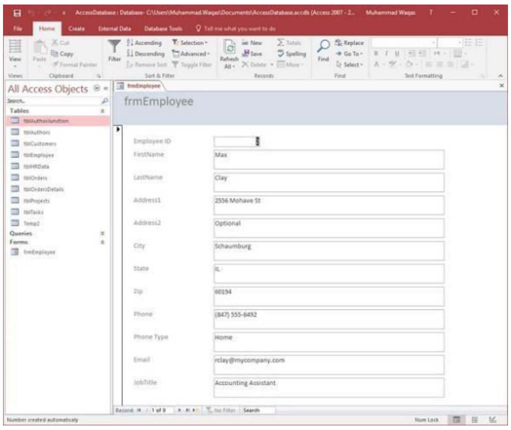
The following screen in the Form Wizard will ask for the layout that we would like forour form. We have **columnar, tabular, datasheet and justified** layouts. We will choose the columnar layout here and then click **Next**.



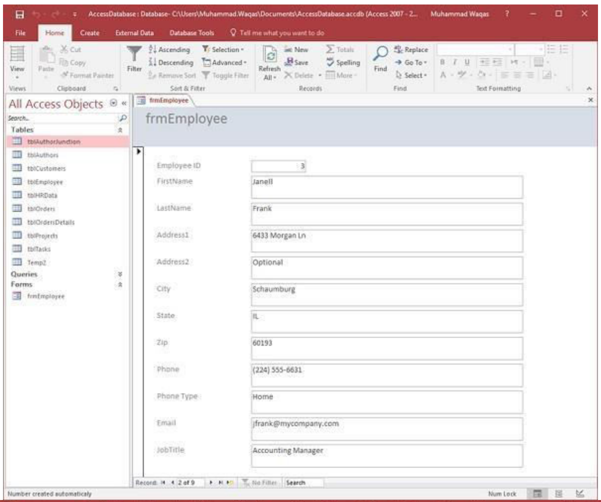
In the following screen, we need to give a title for our form. Let us callit **frmEmployees**.

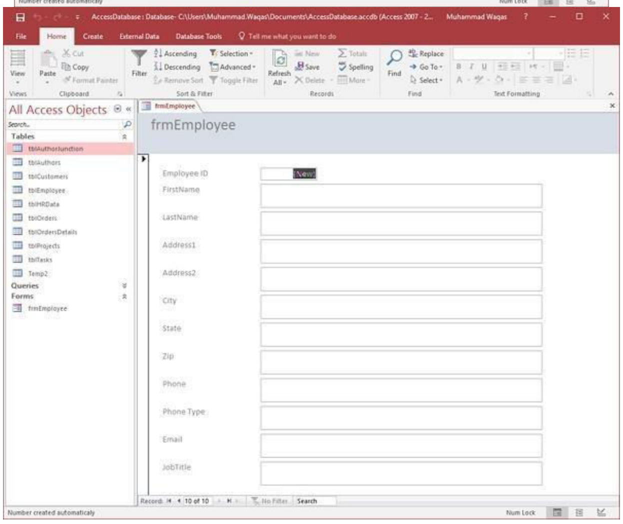


Now, take a look at the following screenshot. This is what your form looks like. This is a single item form, meaning one record is displayed at a time and further down youcan see the navigation buttons, which is telling us that this is displaying the record 1of 9. If you click on that button then, it will move to the next record.



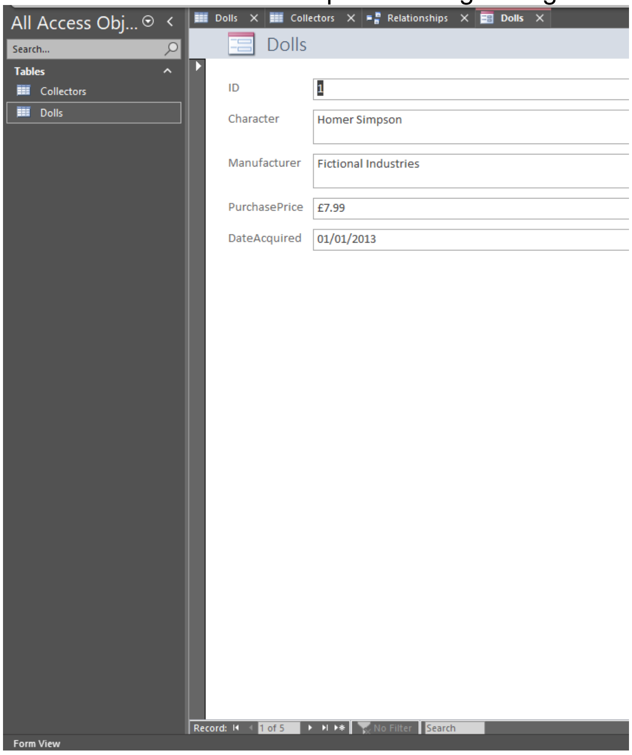
If you want to jump to the very last record in that form or that table, you can use the button right beside that right arrow, the arrow with a line after it, that's the last recordbutton. If you want to add new employee information, go to the end of this records and then after 9 records you will see a blank form where you can begin entering out the new employee's information.





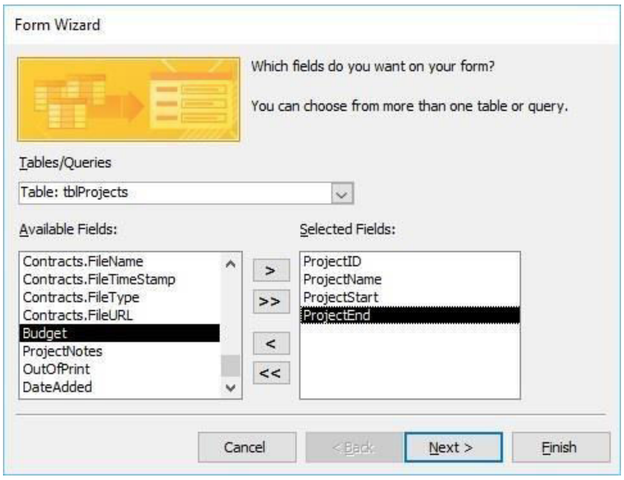
This is one example of how you can create a form using the Form Wizard. Let us nowclose this form and go to the Create tab. After you are done making a form you can edit it’s layout by going into the Layout mode or its design by going into the Design View from the bottom right of your window or from the **Views** Section in the **Home Layout** on top. The Design View gives you much more detailed aspects of the structure(Header, Footer).

Now forms can also be use to insert records or edit already available data in a Table. You can toggle through each record from the arrow keys at the bottom left of the form. You can also search for a specific string throughout the records of the Table.

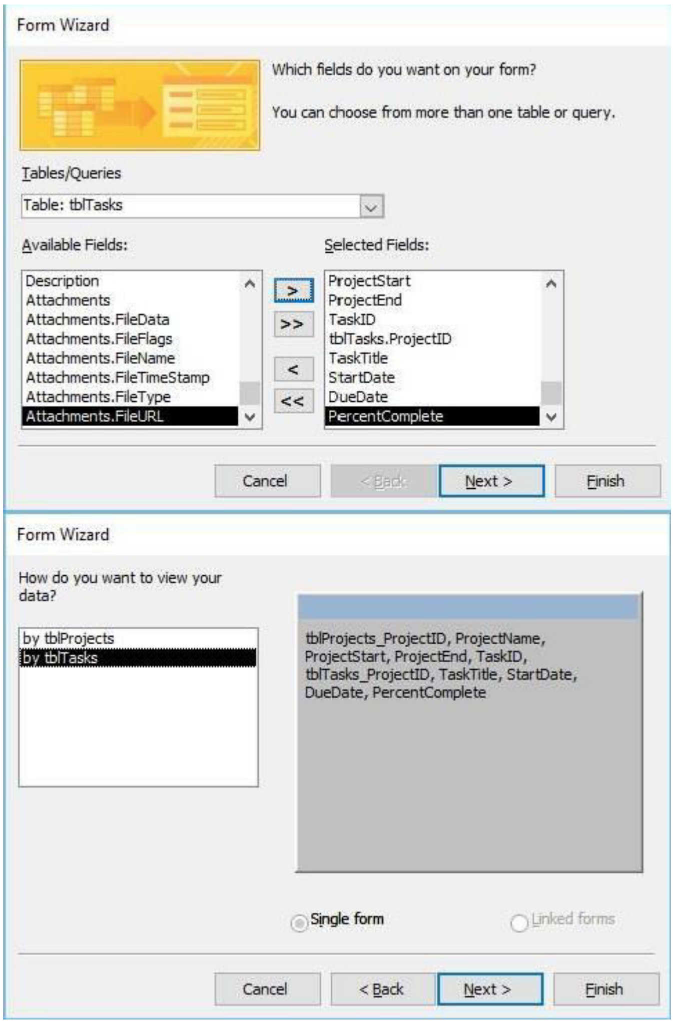


Now we will create a slightly more complicated form using Wizard. Click the Form Wizard and this time, we will choosefields from a couple of different tables

In this Form Wizard, let us choose **tblProjects** for **Tables/Queries**, and select a fewAvailable Fields such as ProjectID, ProjectName, ProjectStart, and ProjectEnd.These fields will now move to Selected Fields.

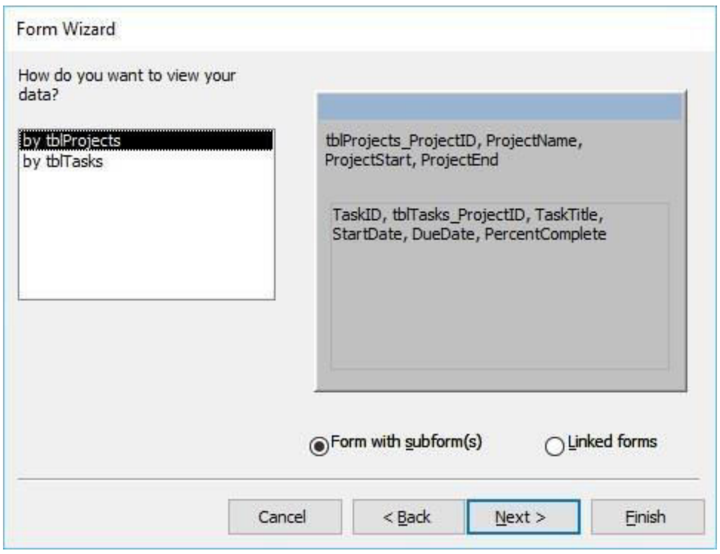


Now select **tblTasks** for Tables/Queries and send over the TaskID, ProjectID, TaskTitle, StartDate, DueDate and PercentComplete. Click **Next**.



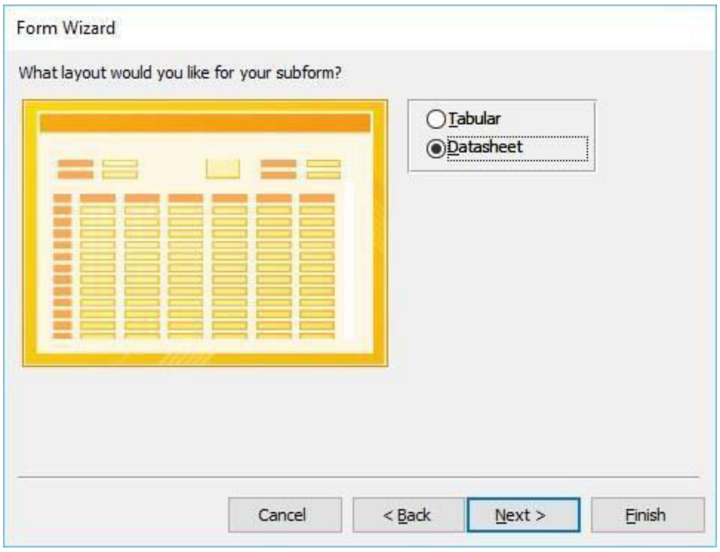
Here, we want to retrieve data from a couple of different objects. We can also choosefrom options on how we want to arrange our form. If we want to create a flat form, we can choose to arrange by **tblTasks**, which will create that single form, with all the fields laid out in flat view as shown above.

However, if we want to create a hierarchical form based on that one-to-many relationship, we can choose to arrange our data by tblProjects.

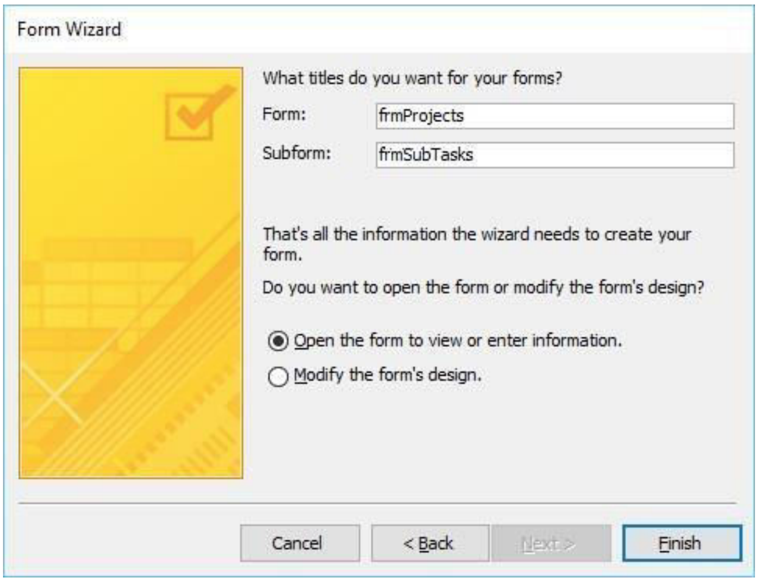


In the above window, we have the option to include a **subform** for **tblTasks**, or we can make that a linked form. This linked form is where tblProjects will have a button that will launch that second form filtered to the project that we have selected in that underlying projects form. Let us now select the **Form with subform(s)**, and then click **Next**.

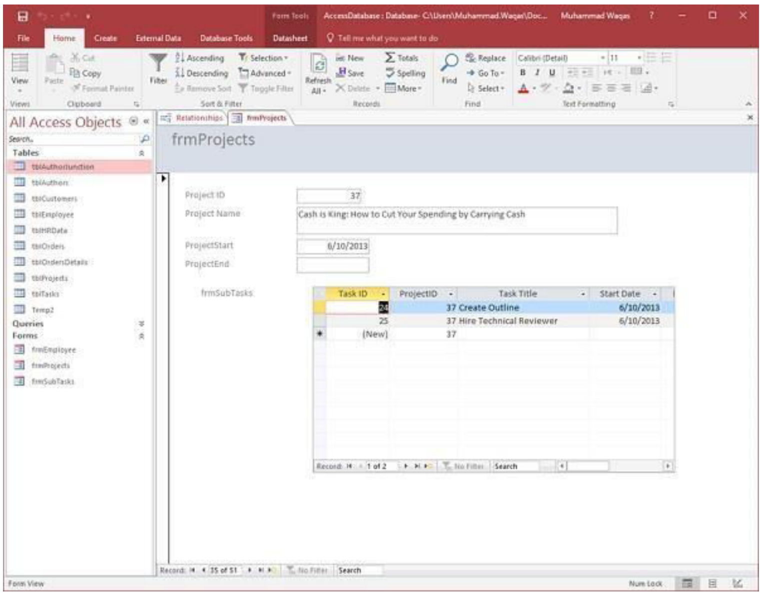
In the following screen, you can choose a layout for your subform. The Datasheet View gets selected by default. The Datasheet View is similar to Table View. Now, click **Next**.



In the following screen, you need to provide a name for your forms. Enter the name you want and click **Finish**.

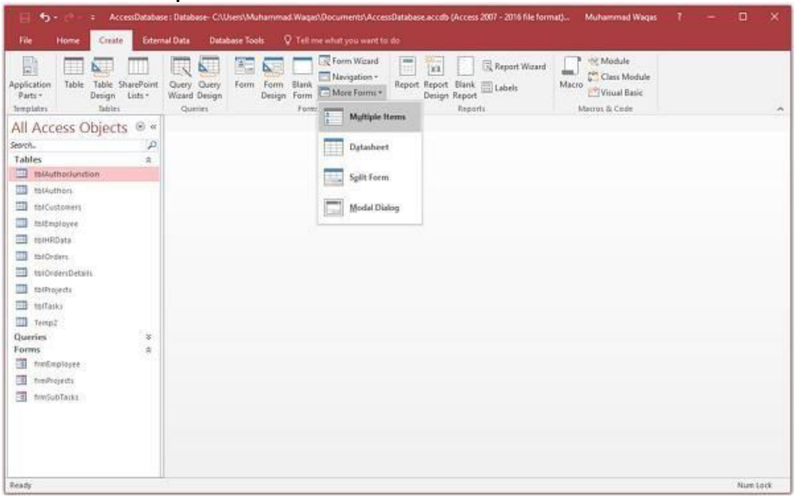


Access will give you a preview of what your form looks like. On top, you have the controls on your main form, which is from our **Projects** table. As you go down, you will see a subform. It's like a form within a form.



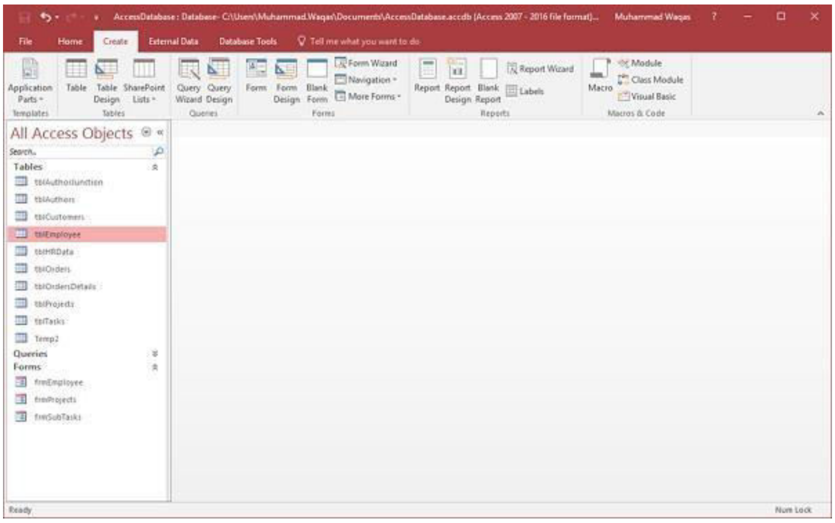
Multiple Item Form

You may also want to create a specific kind of form. For this, you can click onthe **More Forms** drop-down menu.

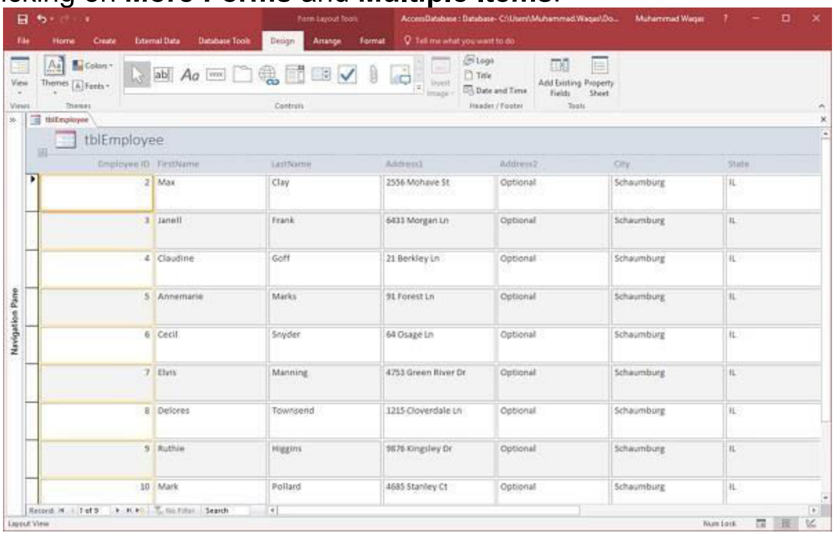


From the menu, you can create a **Multiple Items** form, a **Datasheet** form,a **Split** form, or even a **Modal Dialog** form. These are typically bound forms; select the object that you would like to be bound to that form. This does not apply to the Modal Dialog forms.

To create this type of form, you will need to select the object in navigation pane first.Let us select **tblEmployees** here.



Proceed by clicking on **More Forms** and **Multiple Items**.

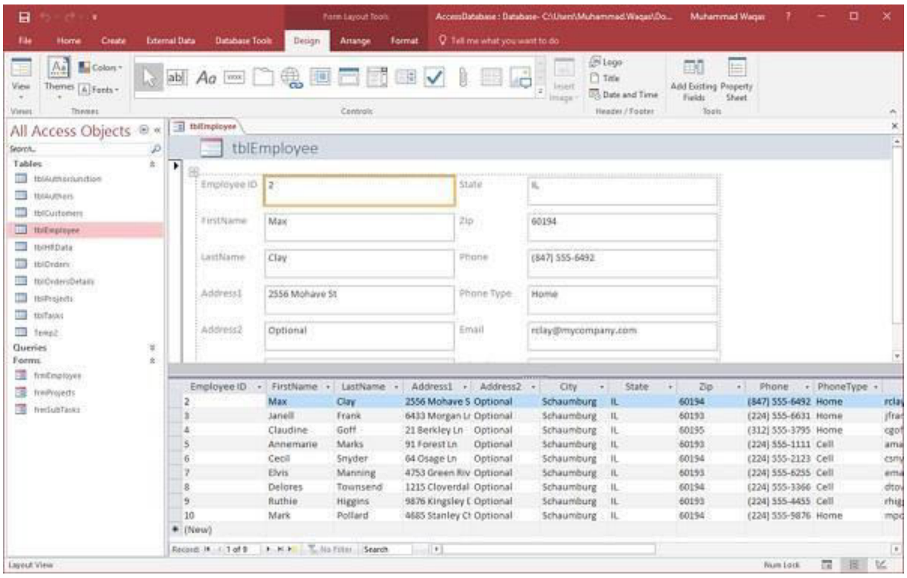


The above step will further create a Multiple Items form, listing out all the employees.

Split Form

This type of form is divided in equal halves, either vertically or horizontally. One half displays a single item or record, and the other half displays a list or a datasheet viewof multiple records from the underlying data source.

Let us now select **tblEmployees** in the navigation pane and then on **Create** tab. Select **Split Form** option from More Forms menu and you will see the following formin which the form is divided vertically.



Form Layout Tools
AccessDatabase : Database- C:\Users\Muhammad.Waqas\Do...
Muhammad Waqas
File
Home
Create
External Data
Database Tools
Design
Arrange
Format
Tell me what you want to do
View
Themes
Colors
Fonts
Views
Themes
Controls
Insert Image
Logo
Title
Date and Time
Header / Footer
Add Existing Fields
Property Sheet
Tools
All Access Objects
Search...
Tables
tblCustomers
tblHRData
tblOrders
tblOrdersDetails
tblProjects
tblTasks
Temp2
Queries
Forms
frmEmployee
frmProjects
frmSubTasks
tblEmployee
Employee ID
2
State
IL
FirstName
Max
Zip
60194
LastName
Clay
Phone
(847) 555-6492
Address1
2556 Mohave St
Phone Type
Home
Address2
Optional
Email
rclay@mycompany.com
Employee ID | FirstName | LastName | Address1 | Address2 | City | State | Zip | Phone | PhoneType
2 | Max | Clay | 2556 Mohave S | Optional | Schaumburg | IL | 60194 | (847) 555-6492 | Home
3 | Janell | Frank | 6433 Morgan Ln | Optional | Schaumburg | IL | 60193 | (224) 555-6631 | Home
4 | Claudine | Goff | 21 Berkley Ln | Optional | Schaumburg | IL | 60195 | (312) 555-3795 | Home
5 | Annemarie | Marks | 91 Forest Ln | Optional | Schaumburg | IL | 60193 | (224) 555-1111 | Cell
6 | Cecil | Snyder | 64 Osage Ln | Optional | Schaumburg | IL | 60194 | (224) 555-2123 | Cell
7 | Elvis | Manning | 4753 Green Riv | Optional | Schaumburg | IL | 60193 | (224) 555-6255 | Cell
8 | Delores | Townsend | 1215 Cloverdal | Optional | Schaumburg | IL | 60194 | (224) 555-3366 | Cell
9 | Ruthie | Higgins | 9876 Kingsley | Optional | Schaumburg | IL | 60193 | (224) 555-4455 | Cell
10 | Mark | Pollard | 4685 Stanley Ct | Optional | Schaumburg | IL | 60194 | (224) 555-9876 | Home
(New)
Record: 1 of 9
No Filter
Search
Layout View
Num Lock